

Section Two: Calculator-assumed

65% (98 Marks)

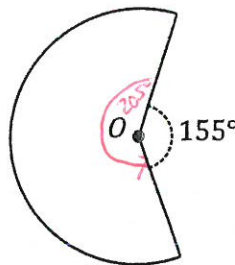
This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9

(5 marks)

A sector of a circle of radius 45 cm is shown below.



- (a) Show that the perimeter of the sector is 251 cm, when rounded to the nearest cm.

(3 marks)

*Sector is within solid line
Must add 2 radii.
Must show full working*

$$\text{Interior Angle} = 360 - 155 \\ = 205^\circ$$

- calculates interior angle ✓

$$\text{Arc Length} = 2\pi(45) \times \left(\frac{205}{360}\right) \\ = 161.0666..$$

- calculates arc length ✓

$$\text{Perimeter} = 161 + 45 + 45 \\ = 251 \text{ cm}$$

- calculates total perimeter ✓

- (b) Determine the area of the sector.

(2 marks)

$$\text{Area} = \pi(45)^2 \times \left(\frac{205}{360}\right) \\ = 3622.649... \text{ cm}^2 \\ \approx 3623 \text{ cm}^2$$

- calculates area of circle ✓

- calculates area of sector ✓

Question 10

(8 marks)

- (a) A triangle has sides of length 55 cm, 48 cm and 71 cm. State, with reasons, whether the triangle is right-angled. (2 marks)

- Substitutes into Pythagoras theorem ✓

$$71^2 = 5041$$

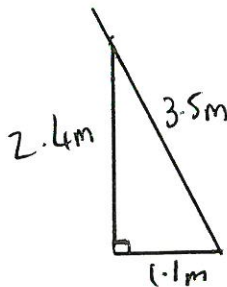
- Explanation that uses the words 'Pythagoras Theorem' ✓

$$48^2 + 55^2 = 5329$$

$5041 \neq 5329$ Not right angled as side lengths do not satisfy Pythagoras theorem.

- (b) A 3.5 m long ladder leans against a 2.4 m tall wall with the foot of the ladder 1.1 m away from the base of the wall on level ground.

- (i) Draw a sketch of this situation, showing the ladder extending beyond the top of the wall. (2 marks)



Many missed right angle or units on diagram.

- Sketch with right angle ✓
- Sketch includes measurements ✓

- (ii) Calculate the distance the ladder extends beyond the top of the wall. (2 marks)

$$d = \sqrt{1.1^2 + 2.4^2}$$

$$= 2.64$$

Must calculate section of ladder that is part of right triangle.

- Calculates length touching wall. ✓

$$\text{Extended} = 3.5 - 2.64$$

$$\text{Distance} = 0.86 \text{ m}$$

- Calculates distance extended above wall. ✓

- (iii) The foot of the ladder is moved along the ground and away from the base of the wall until the top of the ladder is just touching the top of the wall. Determine how far the foot of the ladder was moved. (2 marks)

$$\sqrt{3.5^2 - 2.4^2}$$

- Calculates distance from wall. ✓

Distance from wall

$$= 2.55 \text{ m}$$

$$\therefore \text{Moved } 2.55 - 1.1$$

$$= 1.45 \text{ m}$$

- Calculates the distance ladder moved ✓

Question 11

(7 marks)

- (a) Calculate the value of f given that $f = \frac{1}{d}$ and $d = \frac{1}{d_1} + \frac{1}{d_2}$ when $d_1 = 0.52$ and $d_2 = 2.86$.

(2 marks)

Classpad use
needs improvement.

$$d = \frac{1}{0.52} + \frac{1}{2.86}$$

$$= 2.27$$

$$f = \frac{1}{2.27}$$

$$= 0.44$$

- Calculates correct
value for d ✓

- calculates correct
value for f to 2 D.P ✓

- (b) The future value, F , of an annuity can be calculated using the formula below.

$$F = \frac{C}{r} \left(\left(1 + \frac{r}{n} \right)^{nt} - 1 \right)$$

Consider an annuity with values $C = \$2\,500$, $r = 0.035$, $n = 4$ and $t = 5$.

- (i) Calculate the future value, rounding your answer to the nearest dollar. (3 marks)

over Classpad
practice was
obvious.

$$F = \frac{2500}{0.035} \left(\left(1 + \frac{0.035}{4} \right)^{20} - 1 \right)$$

$$= 13\,595.699...$$

$$F \approx \$13\,596$$

- Substitutes
correctly ✓

- calculates F ✓

- Rounded to the
nearest dollar ✓

- (ii) Calculate the change in F when the value of n is changed from 4 to 12, rounding your answer to the nearest dollar. (2 marks)

Many students did
not calculate change

New value for $F \approx \$13\,639$ (to the nearest dollar)

Students should not
be using rounded values
from part i).
Use Classpad more
effectively.

Therefore increase of $13\,639 - 13\,596$

$$= \$43$$

- Calculates new F ✓

- calculates increase,
correct to the
nearest dollar ✓

Question 12

(9 marks)

The weekly sales of Waywoo phones (the Nine, Plus and Deluxe models) at retail outlets A, B and C is shown in matrix M .

$$M = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} \text{Nine} \\ \text{Plus} \\ \text{Deluxe} \end{matrix} & \begin{bmatrix} 70 & 35 & 41 \\ 45 & 25 & 32 \\ 44 & 36 & 31 \end{bmatrix} \end{matrix}$$

- (a) In total, how many phones were sold by store C?

(1 mark)

$$41 + 32 + 31 = 104$$

- calculates correct number of phones ✓

- (b) What does the element m_{13} in matrix M represent?

(1 mark)

The number of Nine models sold at outlet C.

- Mentions store and model ✓

* must state "sold"

- (c) Determine matrix F , where $F = M \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and explain what F shows.

(2 marks)

* Did not accept "brand".

$$F = \begin{matrix} \text{No. Sales} \\ 9 \\ + \\ \text{oe} \end{matrix} \begin{bmatrix} 146 \\ 102 \\ 111 \end{bmatrix}$$

- correct matrix F ✓

- appropriate explanation ✓

* "Each" is the important word to use

Matrix F shows the total number of each model sold during the week.

* Accepted different phone models [correct context]

- (d) Determine the matrix G , where $G = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \times M$, and explain what G shows. (2 marks)

$$G = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} 1 & 1 & 1 \end{matrix} & \begin{bmatrix} 159 & 96 & 104 \end{bmatrix} \end{matrix}$$

correct matrix G

- appropriate explanation ✓

The total number of phones sold at each of the outlets during the week.

- (e) The Nine, Plus and Deluxe models are sold for \$749, \$899 and \$979 respectively.
- (i) Use this information to write matrix H that can be multiplied by matrix F to yield a sensible result. (1 mark)

$$H = \begin{bmatrix} 749 & 899 & 979 \end{bmatrix}$$

- correct matrix H ✓

- (ii) Multiply matrices H and F together and explain what the result shows. (2 marks)

$$H \times F = \begin{bmatrix} 309721 \end{bmatrix}$$

- The explanation must be on

The context of the store, i.e. revenue, earnings, total amount of money received by the stores etc.

- Cannot be total price paid, profit etc.

The total revenue from selling all the phones at all the stores.

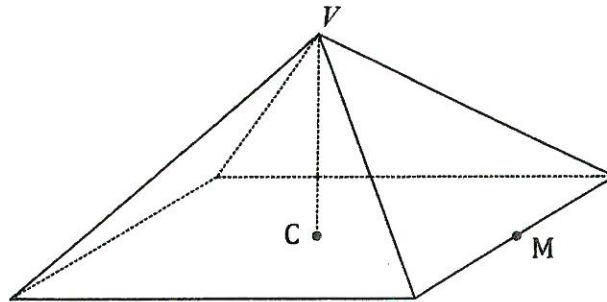
- correct product as a matrix ✓

- appropriate explanation ✓

Question 13

(6 marks)

A bakery sells a specialty cake in the shape of a square-based pyramid. The sides of the square are all 36 cm and the vertex of the pyramid, V , lies 25 cm directly above the centre of the square base, C . The mid-point of one of the sides of the square is M .



- (a) Determine the volume of the cake.

(2 marks)

$$V = \frac{1}{3} \times 36^2 \times 25$$

$$= 10800 \text{ cm}^3$$

- substitutes into volume formula ✓
- calculates correct volume ✓

- (b) Calculate the length VM .

(2 marks)

$$MC = 36 \div 2$$

$$= 18$$

$$VM = \sqrt{25^2 + 18^2}$$

$$= 30.81 \text{ cm}$$

- Shows use of correct lengths for MC and VC ✓
- Calculates correct distance for VM ✓

- (c) The four triangular faces are iced at a cost of 1.1 cents per square cm. Determine the cost of icing the cake, in dollars and cents.

(2 marks)

1 mark awarded if rounded measurement was used to find area.

$$\text{Area of faces} = 4 \times \left(\frac{1}{2} \times 36 \times 30.81 \right)$$

$$= 2218.02 \text{ cm}^2$$

$$\text{Cost} = 2218.02 \times 0.011$$

$$= \$24.40$$

- Calculates correct total surface area ✓
- Calculates correct cost in dollars and cents ✓

Follow through allowed for cost.

Question 14

(6 marks)

A researcher at a weather station uses the following formula to calculate A , the absolute humidity of the air, where R is the relative humidity and T is the air temperature.

$$A = \frac{13.1 \times R \times (1.06)^T}{T + 275}$$

- (a) Calculate A when $R = 55$ and $T = 20$.

(2 marks)

$$A = \frac{13.1 \times 55 \times (1.06)^{20}}{20 + 275}$$

$$= 7.83$$

- Substitutes correctly into formula

- Calculates A

- (b) Calculate R when $A = 8$ and $T = 28$.

(2 marks)

$$8 = \frac{13.1 \times R \times (1.06)^{28}}{28 + 275}$$

- Substitutes correctly into formula

- Solves for R

Solve
classpad

$$R = 36.20$$

The formula for A was used to create the spreadsheet below for various values of R and T .

A		R				
		30	40	50	60	70
T	-5	1.09	P	1.81	2.18	2.54
	0	1.43	1.91	2.38	2.86	Q
	5	1.88	2.50	3.13	3.76	4.38
	10	3.35	3.29	4.12	4.94	5.76

- (c) Calculate the value of the entries P and Q shown in the spreadsheet.

(2 marks)

$$P = 1.45$$

$$Q = 3.33$$

- Calculates correct value for P

- Calculates correct value for Q

For Q ,
if the student
give the answer
as 3.34 X
but if they give
the answer as
3.3345
and
round to 3.34,

-1 overall for rounding

See next page

Question 15

(12 marks)

A spreadsheet to track a monthly car budget is shown below, with all amounts in dollars. The figures for months 1, 2 and 3 show the actual expenditure for each item.

Item	Monthly Budget	Month 1	Month 2	Month 3	3-Month Total
Loan payment	405	405	405	405	1215
Insurance	225	225	225	225	R
Maintenance	108	0	458	45	503
Fuel	162	215	174	Q	582
Total	P	845	1262	858	S

- (a) Determine the values of P, Q, R and S in the spreadsheet.

(3 marks)

$$P = 900$$

$$Q = 193 \quad \text{or} \quad 183$$

$$R = 675$$

$$S = 2965 \quad \text{or} \quad 2975$$

- Calculates one correct value ✓

- Calculates at least three correct values ✓

- Calculates all values correctly ✓

No deduction if they do not include \$.

- (b) A motoring organisation recommends allocating no more than 15% of a person's monthly income to running a car. Would a person on a salary of \$67 500 p.a., using the monthly budget figure above, meet this recommendation? Justify your answer. (3 marks)

Also awarded marks for bringing monthly budget to 1 car and compared with 15% of yearly.

$$\begin{aligned} \$67500 \div 12 \\ = \$5625/\text{month} \end{aligned}$$

$$\begin{aligned} 0.15 \times 5625 \\ = \$843.75 \end{aligned}$$

- Calculates monthly salary ✓

- Calculates correct percentage ✓

- Explains why they do not meet target ✓

∴ Budget of \$900 is over recommended allocation for running a car.

- (c) Over the three months shown, the total actual expenditure for two items exceeded their budgeted figures.

(i) Which items were these?

(1 mark)

- Maintenance
- Fuel

Correct items ✓

- (ii) Determine which of these two items exceeded their budgeted figure by the largest percentage. (3 marks)

$$M = 503 - 108(3) \\ = 179$$

also paid

$$\frac{503}{324} \times 100 = 105\%$$

$$\frac{179}{324} \times 100 = 55.25\%$$

$$F = 582 - 162(3) \\ = 96$$

$$\frac{582}{486} \times 100 = 100\%$$

$$\frac{96}{486} \times 100 = 19.75\%$$

∴ Maintenance exceeded budgeted figure by the largest percentage.

- Calculates % figure for maintenance. ✓

- Calculates % figure for fuel. ✓

- States that maintenance exceeded budget by the largest percentage. ✓

- (d) The monthly budget figure for insurance includes a 10% tax. Determine the cost of the insurance without this tax. (2 marks)

$$225 \div 1.10 \\ = \$204.55$$

- Indicates appropriate method ✓

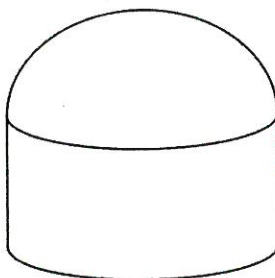
- Calculates correct cost ✓

If they used correct method but incorrect value, awarded 1 mark

Question 16

(6 marks)

A wood turner crafted solid shape S as shown below - a hemisphere of radius 12 cm atop a cylinder with the same radius and height 20 cm.



- (a) The wood turner started out with a block of wood in the shape of a rectangular prism with dimensions $35 \times 30 \times 30$ cm. Determine the volume of wood that was removed from this block to end up with shape S . (3 marks)

$$\begin{aligned}\text{Volume of Prism} &= 35 \times 30 \times 30 \\ &= 31500 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Volume of Cylinder} &= \pi (12)^2 (20) \\ &= 9047.79 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Vol of } \frac{1}{2} \text{ sphere} &= \frac{4}{3} \pi (12)^3 \div 2 \\ &= 3619.11 \text{ cm}^3\end{aligned}$$

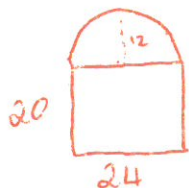
- Calculates volume of cylinder ✓

- Calculates volume of hemisphere ✓

- Calculates volume of removed section ✓

$$\begin{aligned}\text{Volume of Removed Section} &= 31500 - 9047.79 - 3619.11 \\ &= 18833.10 \text{ cm}^3\end{aligned}$$

- (b) Shape S was then cut vertically into two congruent halves. Determine the area of one of the cut faces. (3 marks)



Area of Rectangle

$$\begin{aligned}&= 24 \times 20 \\ &= 480 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of Semi Circle} &= \frac{1}{2} (\pi) (12^2) \\ &= 226.19 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Total area of cut} &= 706.19 \text{ cm}^2\end{aligned}$$

- Uses correct cross sections ✓
+ area of rectangle
- Calculates semi circle area ✓

- Calculates total area of cut ✓

Question 17

(7 marks)

A shop sold three sizes of the same brand of tomato sauce. The 250 mL bottle cost \$1.55, the 475 mL bottle cost \$2.66 and the 920 mL bottle cost \$5.52.

Retail laws require the shop to display the price per 100 mL for each bottle, to the nearest cent.

- (a) Calculate the price per 100 mL for all three sizes of tomato sauce. (3 marks)

$$250 \text{ mL} : \frac{\$1.55}{2.5} = \$0.62 / 100 \text{ mL}$$

$$475 \text{ mL} : \frac{\$2.66}{4.75} = \$0.56 / 100 \text{ mL}$$

$$920 \text{ mL} : \frac{\$5.52}{9.2} = \$0.60 / 100 \text{ mL}$$

- creates an appropriate rate ✓

- all prices correct ✓

- all prices shown as price / 100 mL ✓

Many students lost marks for units (\$) and rounding in this question. 2 DP for money questions!

- (b) Use the unit prices to list the bottle sizes in order of value, from best to worst. (1 mark)

475 mL, 920 mL, 250 mL

Correct order ✓

Must list bottle sizes.

- (c) Determine the price, to the nearest cent, for a 1.25 L bottle of the same sauce if it turned out to be exactly 2 cents per 100 mL cheaper than the best value bottle. (2 marks)

$$\text{Price for 1.25 L Bottle} = \$0.54 \times 12.5 = \$6.75$$

- identifies correct price per 100 mL ✓

- calculates price for the 1.25 L bottle ✓

Well done by many!

- (d) Suggest one reason that a shopper may prefer a size other than that which offers the best value in terms of unit price. (1 mark)

- avoid wastage
- don't have enough money etc.

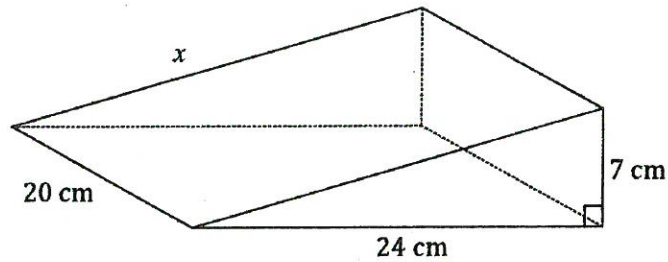
- sensible reason ✓

Be clear with explanations.

Question 18

(8 marks)

A model of a ramp in the shape of a right-triangular prism has dimensions shown in the diagram below.



- (a) Calculate x , the length of the sloping face.

(2 marks)

$$\begin{aligned} x^2 &= 7^2 + 24^2 \\ x &= \sqrt{625} \\ &= 25 \text{ cm} \end{aligned}$$

- Substitutes into Pythagoras Theorem ✓
- Calculates correct length ✓

- (b) Determine the total surface area of the model.

(3 marks)

Rectangles

$$20 \times (7 + 24 + 25) = 1120$$

Triangles

$$2 \times \frac{1}{2} \times 24 \times 7 = 168$$

$$\begin{aligned} \text{Total Surface Area} &= 168 + 1120 \\ &= 1288 \text{ cm}^2 \end{aligned}$$

- Calculates area of all rectangles ✓
- calculates area of triangles ✓
- Calculates TSA ✓

- (c) Determine the volume of the prism.

(1 mark)

$$\begin{aligned} V &= \left(\frac{1}{2} \times 24 \times 7 \right) (20) \\ &= 1680 \text{ cm}^3 \end{aligned}$$

- Calculates volume ✓

- (d) The actual ramp is to be made from concrete that costs \$75 per cubic metre. Given that the volume of the actual ramp is 216 times the volume of the model, determine the cost of concrete for the actual ramp.

(2 marks)

$$\begin{aligned} V &= 1680 \times 216 \\ &= 362880 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Cost} &= 362880 \div 100^3 \times 75 \\ &= \$27.22 \end{aligned}$$

- Indicates volume in cubic centimetres ✓

- Calculates correct cost ✓

Converting to m^3 by $\div 100^3$ was done very poorly!

See next page

Question 19

(8 marks)

Single people aged between 18 and 21, living at home and looking for work qualify for the youth allowance of \$304.40 per fortnight, unless they earn more than \$668.17 in that time, in which case no allowance is paid.

In any fortnight that they earn between \$143 and \$250, the allowance will be reduced by 50 cents in the dollar for earnings over \$143.

In any fortnight that they earn between \$250 and \$668.17, the allowance will be reduced by \$53.50 plus 60 cents in the dollar for earnings over \$250.

Determine, with justification, the fortnightly youth allowance paid to each of the following single people, all aged between 18 and 21.

(a) Alex, who earns \$66 per week.

(2 marks)

- Full allowance as income/fortnight is $66 \times 2 = \$132$
which is below \$143.

$\therefore$ Allowance Alex will receive \$304.40

- Indicates fortnightly earnings ✓
- States correct allowance ✓

Must state how much Alex will receive for both marks.

(b) Bay, who has a part time job paying \$15.25 per hour for 12 hours each fortnight. (3 marks)

Earnings $15.25 \times 12 = \$183$

$\therefore$ Allowance paid = $304.40 - (183 - 143) \times 0.5$
= \$284.40

- Calculates fortnightly earnings ✓
- Calculates deduction ✓
- Calculates correct allowance paid ✓

(c) Cara, who is paid \$16.40 per hour for working 6 hours a day for 3 days a week.

(3 marks)

Earnings = $16.40 \times 6 \times 3 \times 2$
= \$590.40

Allowance paid = $304.40 - (590.40 - 250) \times 0.6 - 53.50$
= \$46.66

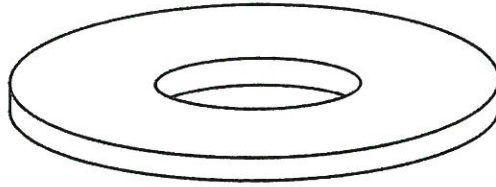
- Calculates fortnightly earnings ✓
- Calculates both parts of reduction ✓
- Calculates correct allowance paid. ✓

Not many allow though is students did not understand conditions of youth allowance.

Question 20

(8 marks)

The washer shown below is made of solid plastic and has an internal radius of 11 mm, an external radius of 28 mm and is 5 mm thick.



→ Must SHOW logical steps → (Messy!)
Vol of washer =
Jumble of numbers
on a page not acceptable.

- (a) The plastic used to make the washers costs \$1.25 per litre. Given that one litre is 1 000 cubic centimetres, determine the cost of making 2 000 of the washers. (4 marks)

→ Main problem 1000 mm³ = 1 cm³. Conversion not completed.

$$\text{Volume of Washer} = (\pi \times 2.8^2 - \pi \times 1.1^2) \times 0.5$$

$$= 10.41 \text{ cm}^3$$

- Applies volume of a cylinder formula ✓

→ Measurements in mm - Easier to convert to cm first.

$$\text{Total Volume} = 10.41 \times 2000$$

$$= 20\,828.76 \text{ cm}^3$$

- Calculates volume of one cylinder ✓

→ Accuracy values must follow through from calculator and rounded once at the end of problem

$$\text{Total cost} = 20\,828.76 / 1000 \times 1.25$$

$$= \$26.04$$

- Volume of all washers ✓

- Calculates correct total cost ✓

BOTH PARTS accuracy poor!

- (b) Calculate the total surface area of one washer, giving your answer in mm². (4 marks)

Needed a systematic method.

$$\text{Area of Base \& Top} = \pi(28)^2 - \pi(11)^2 \times 2$$

$$= 4165.75 \text{ cm}^2$$

- Calculates area of top and bottom ✓

Main problem was subtraction of inner wall.

$$\text{Area of Inner Wall} = 2\pi(11)(5)$$

$$= 345.58 \text{ cm}^2$$

- Calculates area of inner wall ✓

- Calculates area of outer wall ✓

Very generous follow through on this question

$$\text{Area of Outer Wall} = 2\pi(28)(5)$$

$$= 879.65 \text{ cm}^2$$

- Calculates correct total surface area ✓

Question 21

365 DAYS in a Year!

(8 marks)

- (a) An investor pays \$57 250 into a new account offering interest of 2.85% pa, with interest payable every 90 days. The investor closes the account and withdraws the principal and interest just after the first interest payment.

Calculate the amount they withdraw.

$$A = 57250 \left(1 + \frac{0.0285}{\left(\frac{365}{90}\right)}\right) \left(\frac{365}{90}\right) = \$58899.23 \quad (3 \text{ marks})$$

Was FLAT interest but Compound was followed through BUT HARD.

$$\text{Interest} = 57250 \times 0.0285 \times \left(\frac{90}{365}\right) = \$402.32$$

$$\text{Amount Withdrawn} = 57250 + 402.32 = \$57652.32$$

Answer question.

- Substitutes into simple interest formula ✓
- Calculates correct amount of interest ✓
- Calculates correct amount to be withdrawn ✓

- (b) Another person invested \$4 000 with a bank that offered an interest rate of 5.2% p.a., compounded monthly.

- (i) Determine the value of the investment after two years when interest is compounded monthly. $(2 \times 12) \leftarrow$ use brackets on calc (2 marks)

$$4000 \times \left(1 + \frac{0.052}{12}\right)^{24} = \$4437.40$$

- Substitutes correctly into compound interest formula ✓
- Calculates correct value of investment ✓

- (ii) The person was advised that they would be better off after two years if they invested the same sum with another bank that was offering an interest rate of 4.8% p.a. compounded daily. State, with justification, whether this was good advice. (3 marks)

$$4000 \times \left(1 + \frac{0.048}{365}\right)^{(365 \times 2)} \rightarrow \text{use brackets on calc.}$$

$$= \$4403.01$$

\$4403.008465 Rounding

- Substitutes correctly into compound interest formula ✓

- Calculates correct value ✓

∴ Not good advice as earns \$34.40 less interest

- States with appropriate justification that was bad advice. ✓

use correct language from context eg interest value of investment.

Supplementary page

Question number: _____